

INX FUTURE INC. EMPLOYEE PERFORMANCE ANALYSIS

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INTRODUCTION

INX Future Inc. is a leading data analysis and automation solutions provider with a strong global business presence. The company is focused on maintaining a high performance workforce, but recent employee performance indexes has raised concerns among management.

The purpose of this project is to analyze employee performance data and identify the factors associated with employee performance ratings. Exploratory Data Analysis was conducted to understand patterns and relationships within the data. Several machine learning classification algorithms were then trained and evaluated in order to identify the most suitable model for predicting employee performance.

The selected model is used to predict the performance rating of an employee based on relevant employee related factors. The findings from this analysis give insights that can support employee development, performance improvement and hiring decisions at INX Future Inc.

ALGORITHMS AND TRAINING METHODS

Four machine learning classification algorithms were trained and evaluated to determine the most suitable model for predicting employee performance. The algorithms used were Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

Logistic Regression was used as a baseline classification model to establish an initial performance level.

Decision Tree was used to identify patterns in employee characteristics and make predictions through a series of decision rules.

Random Forest was used as a learning method that combines multiple decision trees to improve prediction performance and reduce overfitting.

Gradient Boosting was used as a learning technique that builds models sequentially with each new model attempting to improve the errors made by previous models.

Before training occurred categorical variables were converted into numerical format using one-hot encoding. The “Empnumber” variable was removed because it is an employee identification number and does not provide useful information for predicting performance. The dataset was divided into training and testing sets. The models were evaluated using accuracy, precision, recall and F1-Score. Gradient Boosting achieved the best overall results and was therefore elected as the final model for employee performance prediction.

MODEL TRAINING RESULTS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.50%	81.66%	82.50%	81.77%
Decision Tree	90.83%	90.74%	90.83%	90.76%
Random Forest	91.67%	92.01%	91.67%	91.28%
Gradient Boosting	93.75%	93.75%	93.75%	93.67%

Gradient Boosting achieved the best performance across all four evaluation metrics. It achieved an accuracy of 93.75% and an F1-score of 93.67%. Therefore, Gradient Boosting was selected as the final model for predicting employee performance.

FEATURE SELECTION AND ENGINEERING

Feature selection and engineering were performed to prepare the employee data for machine learning. The “EmpNumber” variable was removed because it is an employee identification number and does not provide meaningful information for predicting performance.

Categorical variables such as Gender, Education background, Marital status, Employee Department, Job Role, Business Travel Frequency, Overtime, and Attrition were converted into numerical variables using one-hot encoding. The “drop-first=True” option was used to avoid unnecessary redundant dummy variables.

Numerical features were retained in their original form. Standard scaling was applied during model preparation, particularly for models that are sensitive to differences in feature scales. The gradient Boosting model selected as the final model was trained using the encoded but unscaled features because tree-based models do not require feature scaling.

IMPORTANT FEATURES AFFECTING EMPLOYEE PERFORMANCE

Feature importance from the selected Gradient Boosting model was analyzed to identify the variables that contributed most to predicting employee performance.

The three most important factors identified were:

- Employee Environment satisfaction
 - Employee Last salary hike percentage
 - Years Since Last Promotion

These results suggest that the employee’s workplace environment, salary growth and opportunities for career progression are important considerations when predicting employee performance. However, feature importance indicates predictive contribution and should not be interpreted as proof that factors directly cause higher or lower performance.

MODEL EVALUATION AND RESULTS

The four classification models were evaluated using accuracy, precision, recall and F1-score. Gradient boosting produced the strongest overall results, achieving an accuracy of 93.75%, precision of 93.75%, recall of 93.75% and an F1-score of 93.67%.

The model correctly classified 225 out of 240 employees in the testing dataset. Based on these results, Gradient Boosting was selected as the final model for employee performance prediction.

The model provides a strong predictive performance and can be used as a decision tool for assessing employee performance based on the available employee related factors.

BUSINESS INSIGHTS

The exploratory data analysis and also the machine learning results provide several insights into the employee performance at INX Future Inc.

Employee performance varies across departments with the work force being concentrated mainly in Sales, Development and Research & Development. Performance Rating 3 represents the main performance level while lower performance ratings require greater attention.

The analysis also identified employee environment satisfaction, employee last salary hike percentage and years since last promotion as the three most important features in the gradient boosting model.

These findings suggest that employee experience within the workplace, financial recognition and opportunities for career progression should receive attention when developing any strategies to improve employee performance.

RECOMMENDATIONS

Based on the findings of this analysis, the following are the recommendations for INX Future Inc.:

- **Improve the workplace environment-** Management should regularly assess employee satisfaction and address workplace issues that may negatively affect employee engagement and also their performance.
- **Review salary growth and recognition-** The company should ensure that salary increases and other forms of recognition are appropriately reward to employee contributions and performance.
- **Provide career development opportunities-** Employees should have clear opportunities for promotion, training, skill development and also career progression.
- **Focus on departments with lower performance-** Management should investigate the causes of lower performance ratings in departments where underperformance is more noticeable and develop targeted improvement strategies.
- **Use the prediction model as a decision-support tool-** The Gradient Boosting model can support management in identifying employees who may require additional development or support. It should be used alongside human judgment rather than as the sole basis for hiring or employee decisions.

TECHNIQUES AND TOOLS USED

The project was developed using Python and several data science and machine learning libraries. Pandas and NumPy were used for data manipulation and numerical operations. Matplotlib and Seaborn were used for exploratory data analysis and for the visualization.

Scikit-learn was used for data preprocessing, model training, and evaluation. The machine learning algorithms used were Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Joblib was used to save and load the trained model and supporting model features for prediction.

Jupyter Notebook was used to organize the data processing, exploratory analysis, model training and prediction stages.

PREDICTION APPLICATION

The trained Gradient Boosting model was saved and used in a separate prediction notebook. Employee information can be provided as input to the model, after which the model processes the information and produces a predicted performance rating.

For example, the test employee used in the prediction notebook received a predicted Performance Rating of 3. This demonstrates how the trained model can be applied to new employee information.

CONCLUSION

This project analyzed employee performance data at INX Future Inc. and developed a machine learning model for predicting employee performance ratings. Exploratory data analysis was used to identify patterns in employee performance, while four classification algorithms were trained and evaluated.

Gradient Boosting achieved the highest overall performance with an accuracy of 93.75% and was then selected as the final prediction model. The feature importance analysis identified Employee Environment Satisfaction, Employee Last Salary Hike Percentage and Years Since Last Promotion as the three most important predictive factors.

These findings can help INX Future Inc. better understand the employee performance and support strategies related to workplace satisfaction, employee recognition and career development. The prediction model can also provide additional decision support for employee development and hiring processes.